Your Proposed Research Title

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For Phase 2, update all black-colored content in this template within your existing Overleaf project.

***Abstract*—[Write your abstract here.]**
***Index Terms*—machine learning, [keyword 2], [keyword 3], [keyword 4]**

I. Introduction

[Introduce the research area and application context.]
[Explain the specific problem and its importance.]
[Briefly summarize existing approaches.]
[State the research gap.]
[Introduce the proposed approach.]
The main contributions of this work are summarized as follows:

- **Contribution 1:** [State the first specific contribution.]
- **Contribution 2:** [State the second specific contribution.]
- **Contribution 3:** [State the third specific contribution.]

II. Related Work

A. Existing Approaches

[Discuss the major categories of existing methods.]

B. Limitations of Existing Work

[Compare the limitations that motivate your proposed study.]

III. Research Challenges and Motivation

A. Challenge 1: [Challenge Name]

Problem: [Precisely define the challenge.]
Evidence from literature: [Cite papers/results demonstrating the problem.]
Importance: [Explain why addressing this challenge matters.]
Technical difficulty: [Explain specifically why this challenge is difficult.]
Connection to proposed method: [Identify the proposed component intended to address it.]

B. Challenge 2: [Challenge Name]

Problem: [Define Challenge 2.]
Evidence from literature: [Provide supporting evidence and citations.]
Importance: [Explain importance.]
Technical difficulty: [Explain technical difficulty.]
Connection to proposed method: [Identify the corresponding proposed technique.]

C. Challenge 3: [Challenge Name]

Problem: [Define Challenge 3.]
Evidence from literature: [Provide supporting evidence and citations.]
Importance: [Explain importance.]
Technical difficulty: [Explain technical difficulty.]
Connection to proposed method: [Identify the corresponding proposed technique.]

IV. Proposed Methodology

A. Problem Formulation

[Formulate the ML problem.]

B. Overall Framework

Insert proposed methodology / ML pipeline diagram here

Fig. 1. Overview of the proposed methodology.

C. Proposed Components

[Describe the proposed components.]

V. Project Execution Plan

A. Six-Week Research and Implementation Plan

VI. Team Member Contributions

- **Student 1:** [Specific contributions.]
- **Student 2:** [Specific contributions.]
- **Student 3:** [Specific contributions.]

TABLE I
COMPARISON OF REPRESENTATIVE RELATED WORKS

Reference	Dataset	Method	Task	Metrics	Key Finding	Limitation / Gap
Paper 1	Dataset	Method	Task	Metric	Finding	Gap
Paper 2	Dataset	Method	Task	Metric	Finding	Gap
Paper 3	Dataset	Method	Task	Metric	Finding	Gap
Paper 4	Dataset	Method	Task	Metric	Finding	Gap
Paper 5	Dataset	Method	Task	Metric	Finding	Gap
....	....	....	....	....	....	

TABLE II
MAPPING BETWEEN RESEARCH CHALLENGES AND PROPOSED METHODS

Challenge	Technique	Function	Expected Effect
Challenge 1	Technique 1	[Function]	[Effect]
Challenge 2	Technique 2	[Function]	[Effect]
Challenge 3	Technique 3	[Function]	[Effect]

VII. EXPERIMENTAL METHODOLOGY

A. Dataset

[Describe the dataset.]

B. Data Preprocessing

[Describe preprocessing.]

C. Data Splitting and Validation

[Describe the validation protocol.]

D. Baseline Methods

- **Baseline 1:** [Method and justification.]
- **Baseline 2:** [Method and justification.]
- **Baseline 3:** [Method and justification.]

E. Implementation Details

[Provide implementation details.]

F. Evaluation Metrics

[Define and justify evaluation metrics.]

VIII. RESULTS AND ANALYSIS

A. Main Results

[Analyze the main results.]

B. Ablation Study

C. Error and Failure Analysis

[Provide error/failure analysis.]

IX. USE OF GENERATIVE AI AND LLM TOOLS

[Provide your disclosure here.]

X. CONCLUSION

[Write the conclusion.]

REFERENCES

TABLE III
SIX-WEEK RESEARCH AND IMPLEMENTATION PLAN EXAMPLE

Week	Objective	Specific Tasks	Expected Output / Milestone	Member(s)
1	Problem and data preparation	Finalize research questions and hypotheses; finalize dataset; implement preprocessing; define train/validation/test protocol; identify evaluation metrics.	Reproducible dataset pipeline; finalized research questions; dataset statistics and experimental protocol.	[Names]
2	Baseline establishment	Implement selected baseline methods; verify implementations; perform initial training and evaluation using the same experimental protocol planned for the proposed method.	Baseline performance table; initial plots; verified baseline implementation.	[Names]
3	Proposed method implementation	Implement the proposed technique/component(s); integrate them with the existing data pipeline; perform debugging and preliminary experiments.	Working implementation of proposed method; preliminary results demonstrating successful execution.	[Names]
4	Main experiments	Train and evaluate the proposed method; perform hyperparameter selection; compare against baselines using the predefined metrics; repeat experiments where necessary.	Main experimental results; comparison tables; performance figures.	[Names]
5	Analysis and validation	Conduct ablation study, error/failure analysis, robustness or sensitivity analysis where appropriate; analyze why the proposed method improves or fails.	Ablation/error analysis; final figures and tables; evidence supporting or rejecting the research hypothesis.	[Names]
6	Finalization and reproducibility	Complete remaining experiments; verify reproducibility; interpret results; identify limitations; finalize manuscript, figures, references, and presentation.	Reproducible final results; completed IEEE-style manuscript; final presentation/poster.	[Names]

TABLE IV
PERFORMANCE COMPARISON

Method	Metric 1	Metric 2	Metric 3
Baseline 1	–	–	–
Baseline 2	–	–	–
Baseline 3	–	–	–
Proposed Method	–	–	–

TABLE V
ABLATION STUDY

Configuration	Primary Metric
Baseline	–
+ Component A	–
+ Component B	–
Full Proposed Method	–